DBMS Lecture 10: Foreign Key & Referential Integrity

Master Study Notes for GATE, PSUs, University Exams & Placements

1 1. Introduction & Core Significance

A **Foreign Key (FK)** is one of the most critical foundational concepts in Relational Database Management Systems (RDBMS). While a Primary Key establishes entity identity within a single table, a Foreign Key establishes cross-table **relationships** and enforces data consistency.

Foreign Key mechanics are heavily tested in:

- **Referential Integrity** rules and violation checks
- **Relational Database Design & Schema Normalization**
- **SQL DDL/DML Statements** (Cascading updates, deletions, constraints)
- **ER Model to Relational Schema Mapping**
- **GATE, ISRO, BARC, NET** and Core Technical Interviews

2 2. Formal Definition of a Foreign Key

Formal Definition

A **Foreign Key** is an attribute or a set of attributes in a relation (referencing relation) that references a candidate key—most commonly the **Primary Key**—of the same or another relation (referenced relation).

Foreign keys are broadly classified by structural complexity:

- **Simple Foreign Key**: Consists of a single attribute referencing a single-attribute primary key.
- **Composite Foreign Key**: Consists of multiple attributes referencing a composite primary key in the referenced table.

3 3. Motivational Example: University Database

Consider two separate tables in a university management system:

Referenced Table: `Student`

Roll_No (PK)	Student_Name	Address
101	Rahul	Bengaluru
102	Ravi	Mysuru
103	Sara	Hubli

Referencing Table: `Course`

Course_ID (PK)	Course_Name	Roll_No (FK)
C01	DBMS	101
C02	Java	102
C03	Python	101

Here, `Course.Roll_No` establishes a logical connection pointing directly to valid entries in `Student.Roll_No`.

4 4. Terminology: Referenced vs. Referencing Table

In relational database design, understanding the structural hierarchy is vital:

Referenced Table (Parent Table)
The table whose candidate key / primary key is being targeted. In our example, **`Student`** is the Referenced Table because it holds the master `Roll_No`.

Referencing Table (Child Table)
The table that contains the foreign key attribute pointing outward. In our example, **`Course`** is the Referencing Table because its `Roll_No` column references the `Student` table.

Parent-Child Relationship Architecture

Referenced Table (Parent) $\xrightarrow{\text{Referenced by}}$ **Referencing Table (Child)**

`Student(Roll_No, Name, Address)` — `Course(Course_ID, Course_Name, Roll_No)`

$\uparrow$ $\downarrow$

Primary Key (PK) $\xleftarrow{\text{Points to}}$ **Foreign Key (FK)**

5 5. Foreign Key & Referential Integrity Rule

A foreign key directly enforces the **Referential Integrity Constraint**.

Referential Integrity Rule

The Referential Integrity Rule states that for any tuple in a referencing relation, the foreign key value must either:

i. Match an existing primary key / candidate key value in the referenced relation, **OR**

ii. Be completely `NULL` (provided the column definition permits null values).

Violation Scenario Example

Given `Student.Roll_No` values $\{101, 102, 103\}$:

Course_ID	Roll_No (FK)	Validation Status
C01	101	**Valid** (101 exists in `Student`)
C02	102	**Valid** (102 exists in `Student`)
C04	999	**Invalid / Rejected** (999 does not exist in `Student`)

6 6. SQL Syntax & Implementation

6.1 Table Creation with Foreign Key

```sql
-- Step 1: Create Parent (Referenced) Table
CREATE TABLE Student (
    Roll_No INT PRIMARY KEY,
    Student_Name VARCHAR(50),
    Address VARCHAR(100)
);

-- Step 2: Create Child (Referencing) Table
CREATE TABLE Course (
    Course_ID VARCHAR(10) PRIMARY KEY,
    Course_Name VARCHAR(50),
    Roll_No INT,
    FOREIGN KEY (Roll_No) REFERENCES Student(Roll_No)
);
```

6.2 Adding Foreign Key to Existing Table

```sql
ALTER TABLE Course
ADD CONSTRAINT FK_Course_Student
FOREIGN KEY (Roll_No) REFERENCES Student(Roll_No);
```

7 7. Essential Characteristics of Foreign Keys

1. **Data Type Compatibility**: The foreign key attribute does not require identical column naming, but its domain and data type **must be compatible** with the referenced key (e.g., `INT` to `INT`).
2. **Name Independence**: Column names can differ across tables. For example:

   $$\texttt{Course.Student\_Roll\_No (FK)} \longrightarrow \texttt{Student.Roll\_No (PK)}$$

3. **Multiple Foreign Keys per Table**: A single relation can contain multiple foreign keys referencing different parent tables.

   ```sql
   CREATE TABLE Orders (
       Order_ID INT PRIMARY KEY,
       Customer_ID INT FOREIGN KEY REFERENCES Customer(Customer_ID),
       Product_ID INT FOREIGN KEY REFERENCES Product(Product_ID),
       Employee_ID INT FOREIGN KEY REFERENCES Employee(Employee_ID)
   );
   ```

4. **Duplicate Values Allowed**: Foreign key columns **do not** have to be unique. Multiple child rows can reference the exact same parent primary key value (e.g., student 101 enrolled in 3 courses).
5. **NULL Values Allowed**: A foreign key column can store `NULL` unless explicitly constrained by `NOT NULL`. A `NULL` FK indicates an unassigned or independent relationship.
6. **Self-Referencing Foreign Key**: A foreign key can reference a primary key within the **same table** (Unary Relationship).

Employee_ID (PK)	Name	Manager_ID (FK)
101	Alice	NULL (Top Executive)
102	Bob	101
103	Charlie	101

7. **Composite Foreign Key**: When referencing a composite primary key, the foreign key must declare the exact corresponding attribute set:

```
FOREIGN KEY (Child_Attr1, Child_Attr2)
    REFERENCES Parent(Parent_Attr1, Parent_Attr2)
```

8 8. Referential Actions: Handling Parent Deletions & Updates

When a row in the parent table is deleted or updated, child tuples holding foreign keys referencing that row would become orphaned. SQL provides explicit referential actions to govern this:

`ON DELETE RESTRICT / NO ACTION` **(Default)**
: Prevents the deletion or modification of a parent row if dependent rows exist in the child table. The DBMS raises a referential integrity violation error.

`ON DELETE CASCADE`
: Automatically deletes or updates all corresponding child rows when the parent row is deleted or updated.

```
FOREIGN KEY (Roll_No) REFERENCES Student(Roll_No)
    ON DELETE CASCADE
    ON UPDATE CASCADE
```

`ON DELETE SET NULL`
: Automatically sets the foreign key values in corresponding child rows to `NULL` when the parent row is deleted (Requires the child column to allow `NULL`).

```
FOREIGN KEY (Roll_No) REFERENCES Student(Roll_No)
    ON DELETE SET NULL
```

`ON DELETE SET DEFAULT`
: Sets the foreign key values in child rows to their default defined value upon parent row deletion.

9 9. Primary Key vs. Foreign Key Comparison

10 10. Master Key Classification Matrix

11 11. Target GATE & Interview Q&A

Q1. What is a foreign key?
: An attribute or set of attributes in a referencing table that targets a candidate key (usually primary key) in a referenced table.

Q2. Which integrity constraint is enforced by a foreign key?
: **Referential Integrity**.

Table 1: In-Depth Technical Comparison Matrix

Feature	Primary Key (PK)	Foreign Key (FK)
Core Purpose	Uniquely identifies tuples within its own table	Establishes logical links to primary/candidate keys
Integrity Rule	Enforces **Entity Integrity**	Enforces **Referential Integrity**
Uniqueness	Must strictly be **UNIQUE**	**Duplicates Permitted** (1-to-N relationships)
NULL Values	**Strictly Forbidden** (`NOT NULL`)	**Allowed** (Unless restricted by `NOT NULL`)
Constraint Count	At most **one** per relation	**Multiple** foreign keys permitted per table
Target Reference	Referenced by foreign keys	References candidate/primary keys
Domain Role	Defines entity identity	Defines inter-entity relationships

Table 2: DBMS Key Types Overview

Key Type	Primary Definition & Functional Role
Super Key	Any attribute set that uniquely identifies every tuple in a relation.
Candidate Key	A minimal super key (contains no redundant attributes).
Primary Key	The primary candidate key selected by the database designer.
Alternate Key	All candidate keys that were not chosen as the primary key.
Foreign Key	An attribute referencing a candidate key of the same/another relation.

Q3. Can a table have multiple foreign keys?
Yes. A table can have as many foreign keys as required by the schema design.

Q4. Can a foreign key contain duplicate values?
Yes. Duplicates allow multiple child records to associate with a single parent record.

Q5. Can a foreign key contain NULL values?
Yes, unless explicitly declared as `NOT NULL`.

Q6. Does a foreign key column name have to match the referenced key name?
No, only the data types and domains must be compatible.

Q7. What is a self-referencing foreign key?
A foreign key that references a candidate/primary key within the same table (e.g., `Manager_ID` referencing `Employee_ID`).

Q8. What is the difference between ON DELETE CASCADE and ON DELETE SET NULL?
`CASCADE` deletes the child rows when the parent row is deleted, whereas `SET NULL` sets the foreign key attribute in the child rows to `NULL`.

Q9. What is the referenced table vs. referencing table?
The referenced table (parent) holds the original primary key; the referencing table (child) holds the foreign key.

Q10. Can a primary key be a foreign key at the same time?
Yes. In weak entity relationships or 1-to-1 table extensions, a primary key column can simultaneously act as a foreign key referencing another table.

12 12. One-Minute Revision & Key Formulas

Ultimate Key Memory Rules

PRIMARY KEY $\Longrightarrow$ **Who am I?** (Entity Identity $\rightarrow$ UNIQUE + NOT NULL)
FOREIGN KEY $\Longrightarrow$ **Where do I belong?** (Relationship $\rightarrow$ Referential Integrity)

Core Rules to Remember for GATE: • Primary Key $\rightarrow$ Entity Integrity
• Foreign Key $\rightarrow$ Referential Integrity • Parent Table = Referenced Table (Holds PK)
• Child Table = Referencing Table (Holds FK)
• FK duplicates allowed: YES | FK NULL allowed: YES (by default)

DBMS Challenge Progress: Lecture 10 Complete · Next Topic: Super Key, Alternate Key & Composite Key Practice Problems